ULX5-PCIe v001
Intergalaktik d.o.o.
License: CERN-OHL-S v2
Year: 2026

power

File: power.kicad_sch

Add RP2040 so that we do not need externa JTAG

Add inline resistors for input signals

ethernet
USB_CC1
USB_CC2
{SD}
{USB}

File: ethernet.kicad_sch

HighSpeed
{DDMI0}
{LVDS0}
{LVDS1}
HDMI_5v
CAM_GPIO0
CAM_GPIO1
SCL1
SCL0
SDA0
SDA1

File: HighSpeed.kicad_sch

File: lvds.kicad_sch
lvds
{DDMI0}
{LVDS0}
{LVDS1}
PERST#
PEWAKE#
CLKREQ#
PEDET
SUSCLK
DEVSLP

PERST#
PEWAKE#
CLKREQ#

PCIexpress_connector
PERST
WAKE
CLKREQ#
TRST
{JTAG}
SMCLK
SMDAT
{PCIe}

File: PCIexpress_connector.kicad_sch

Raspberry_Pico
{PI_JTAG}

File: raspberry_pico.kicad_sch

config
{PI_JTAG}
{JTAG}

File: config.kicad_sch

gpdi-ram
CAM_GPIO0
CAM_GPIO1
SCL0
SDA0

File: ram.kicad_sch

serdes
{PCIe}

File: serdes.kicad_sch

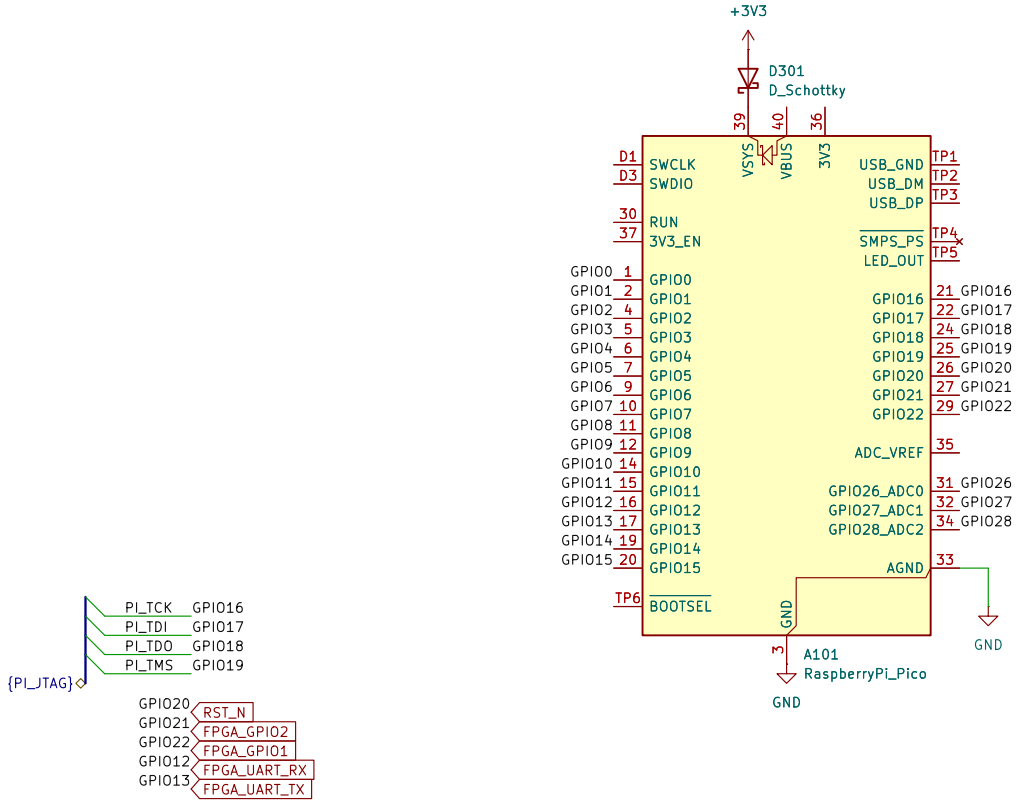
FID104 Fiducial
FID105 Fiducial
FID106 Fiducial
FID101 Fiducial
FID102 Fiducial
FID103 Fiducial

H301 MountingHole_Pad GND
H302 MountingHole_Pad GND
H303 MountingHole_Pad GND
H304 MountingHole_Pad GND
H305 MountingHole_Pad GND
H306 MountingHole_Pad GND
H307 MountingHole_Pad GND

CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /
File: ulx5-PCIe-1x.kicad_sch

Title: ULX5-PCIe

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 1/11



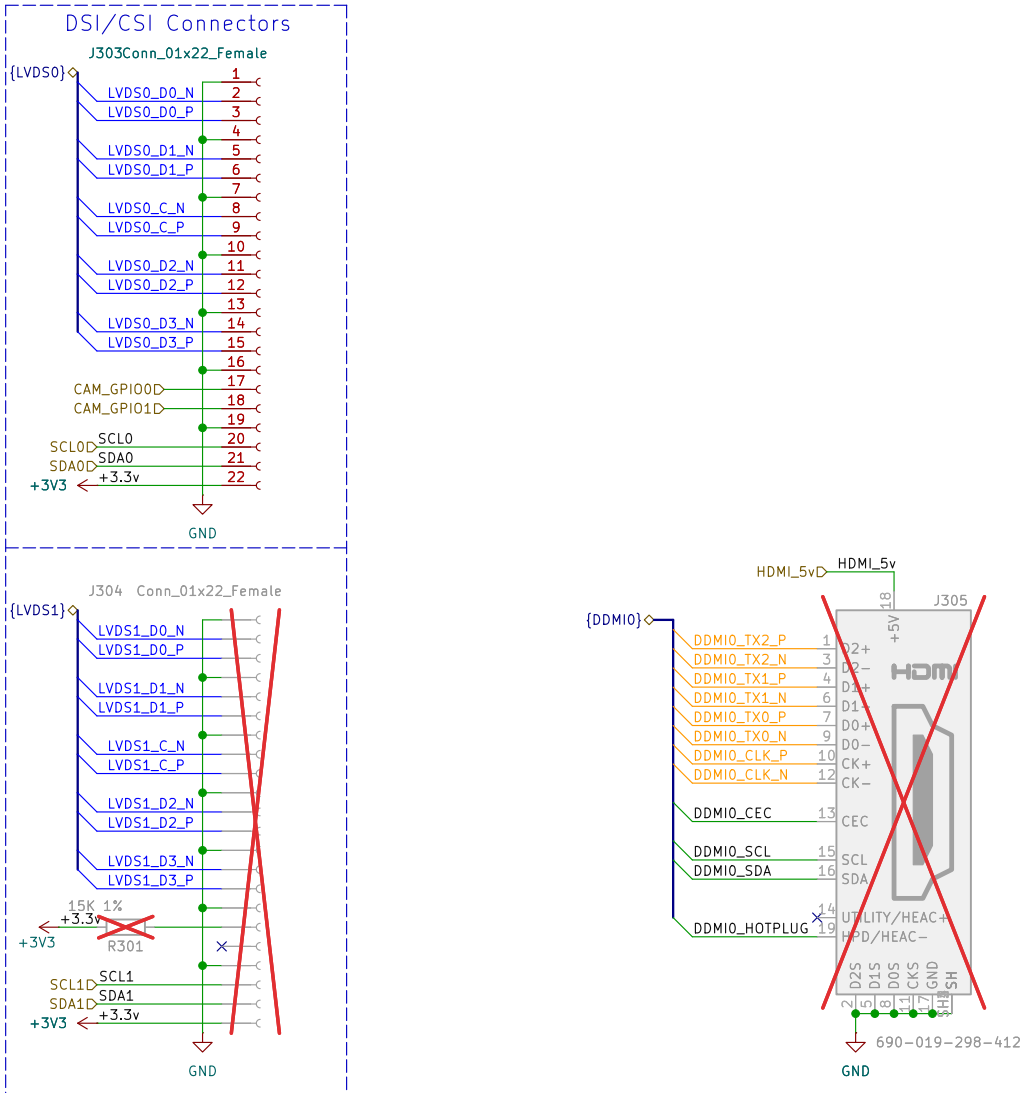
CERN-OHL-S v2

Intergalaktik d.o.o.

Sheet: /Raspberry_Pico/
File: raspberry_pico.kicad_sch

Title: ULX5-PCIe

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 2/11



CERN−OHL−S v2

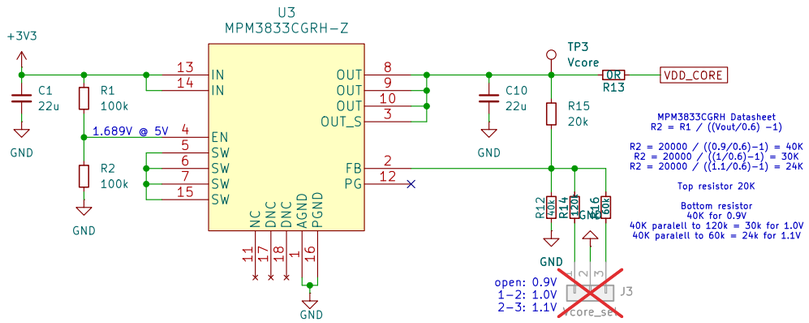
Intergalaktik d.o.o.

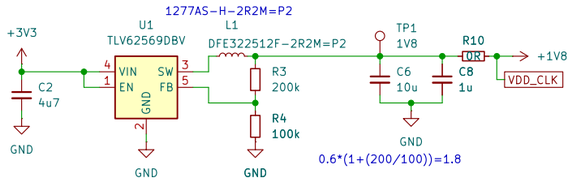
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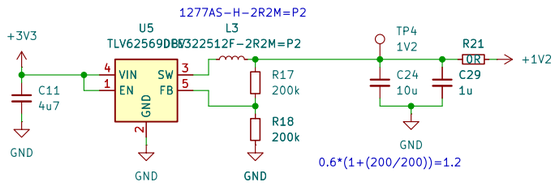
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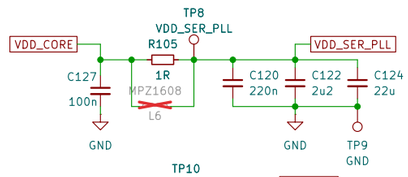
Title: ULX5−PCIe

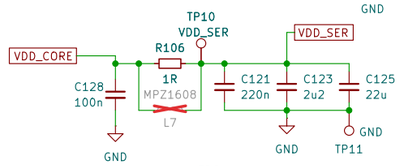
Size: A4	Date: 2025−06−24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 3/11

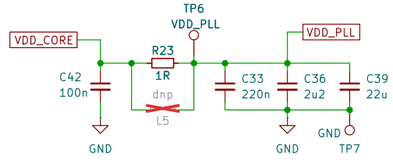


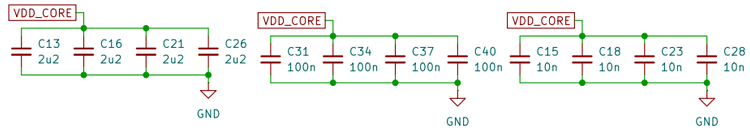


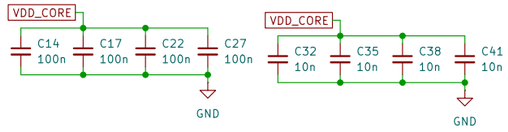


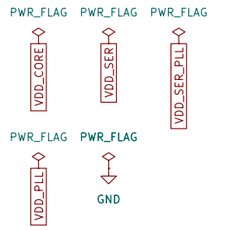


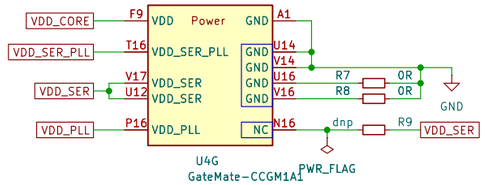


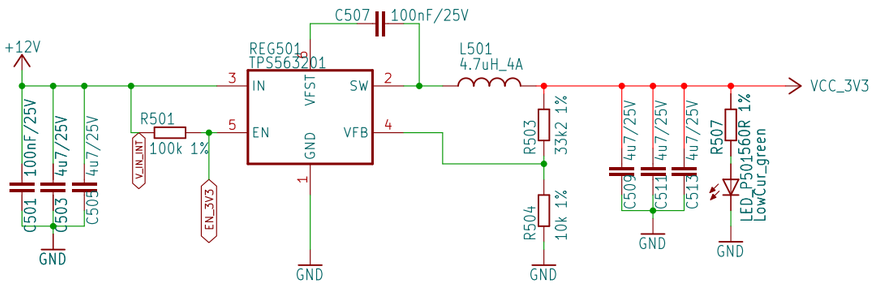


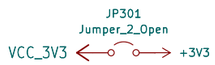












CERN-OHL-S v2		
Intergalaktik d.o.o.		
Sheet: /power/		
File: power.kicad_sch		
Title: ULX5-PCIe		
Size: A3	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 4/11



Use MD0/1 PU/PD for CPHA/CPOL selection

SW1 219-2MST

U4B GateMate-CCGM1A1

U6 74LVC1G04GM,115

U7 74LVC2G00HK3-7

U8 74LVC2G00HK3-7

U9 AT25QL321-MBUE-T

J4 JTAG

J101 GND JTAG_EXT

J102 3V3_PSU

{JTAG}

{PI_JTAG}

CERN-OHL-S v2

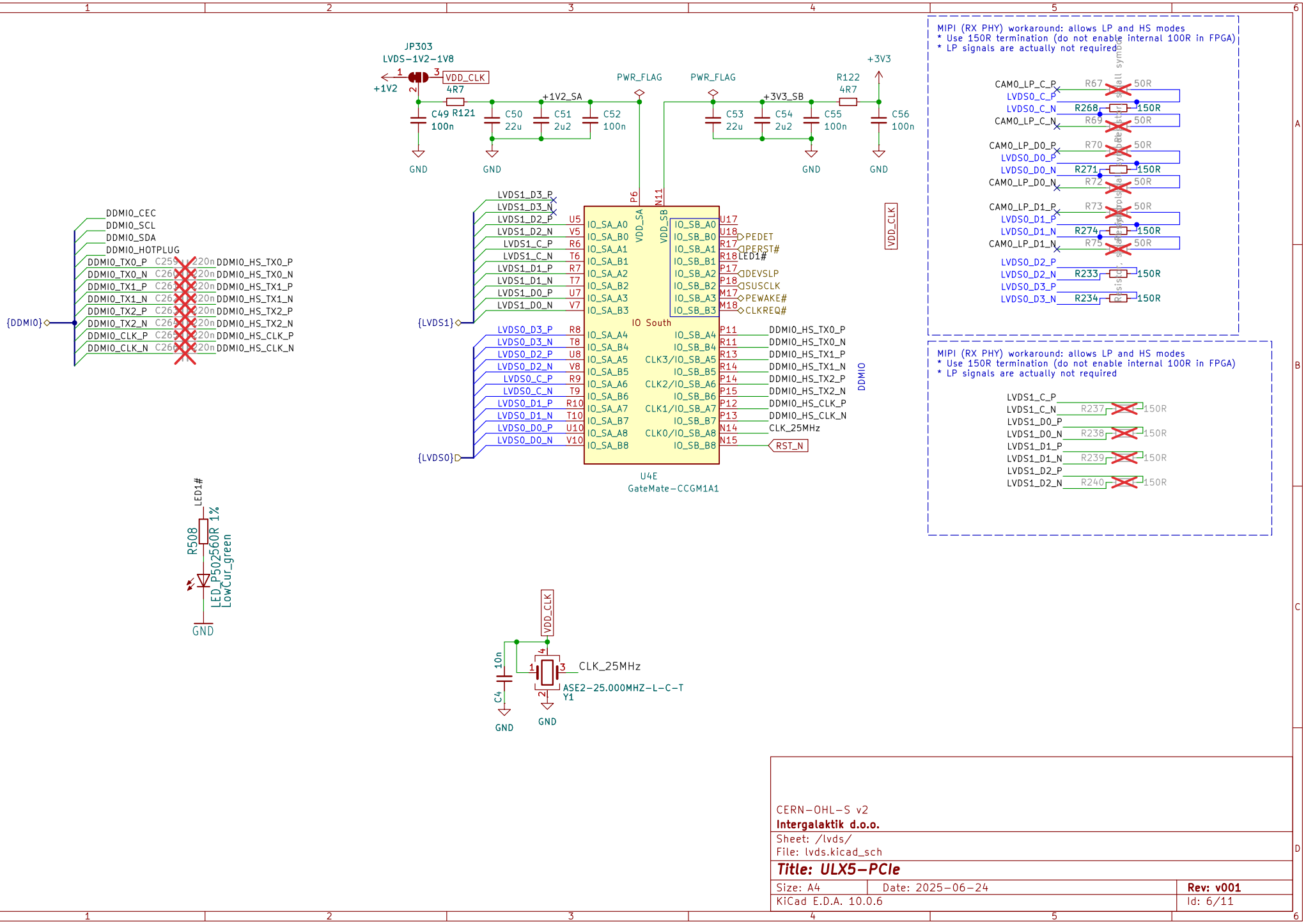
Intergalaktik d.o.o.

Sheet: /config/

File: config.kicad_sch

Title: ULX5-PCIe

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 5/11

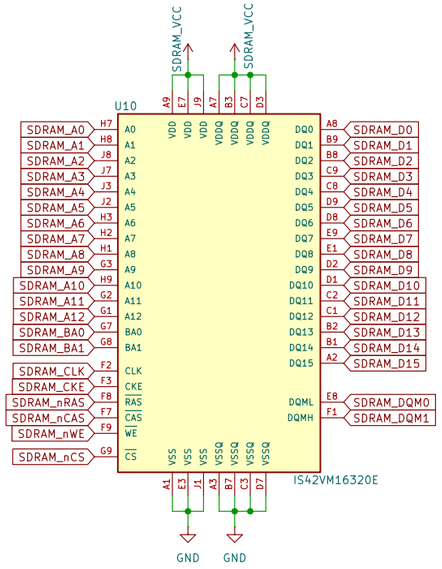


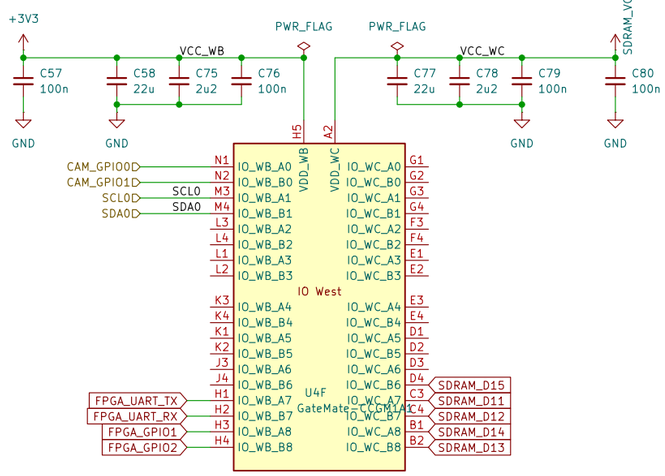
CERN-OHL-S v2
Intergalaktik d.o.o.

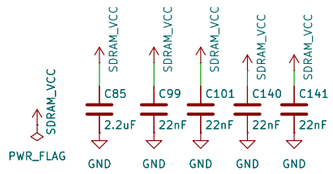
Sheet: /lvds/
File: lvds.kicad_sch

Title: ULX5-PCIe

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 6/11

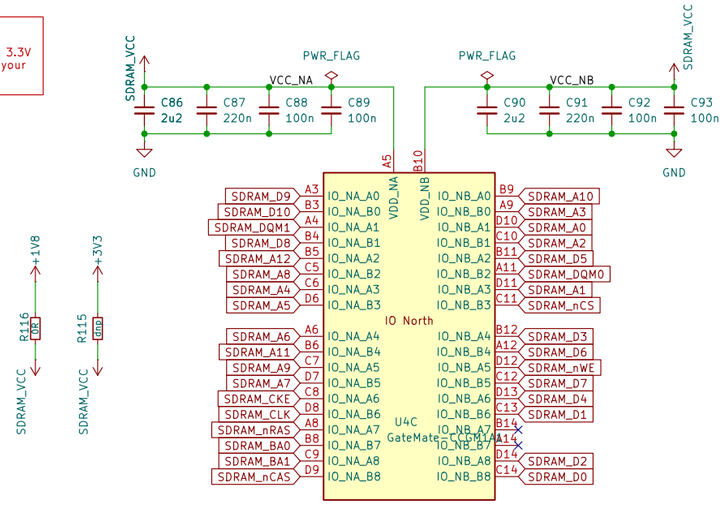






Caution!

GPIO banks are actually not 3.3V tolerant. Use this option at your own risk.



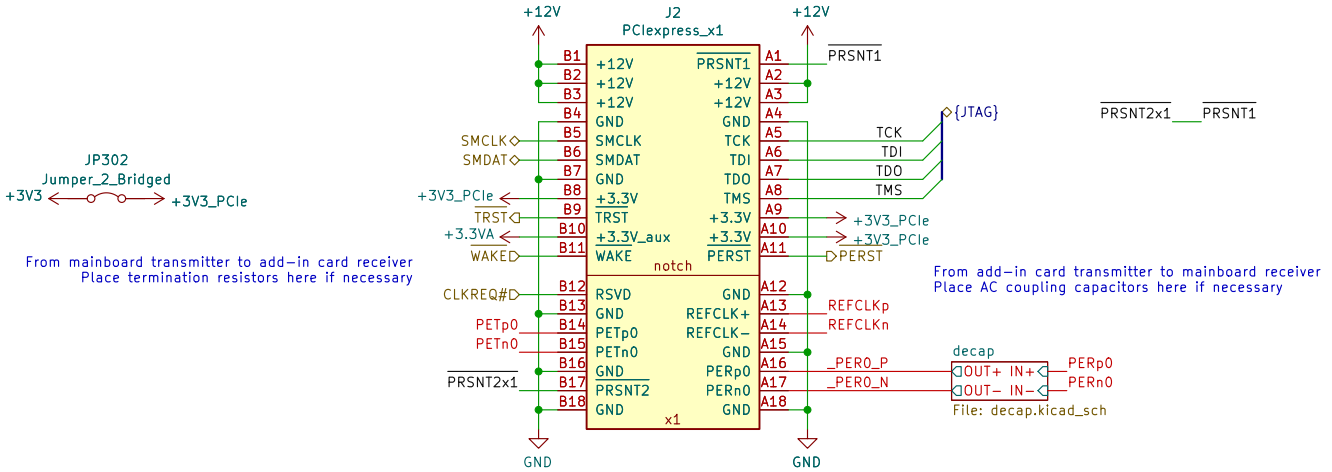
CERN-OHL-S v2

Intergalaktik d.o.o.

Sheet: /gpdi-ram/
File: ram.kicad_sch

Title: ULX5-PCIe

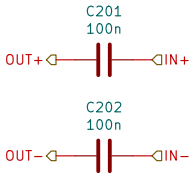
Size: A3	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 7/11



CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /PCIexpress_connector/
File: PCIexpress_connector.kicad_sch

Title: ULX5-PCIe

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 8/11



CERN-OHL-S v2

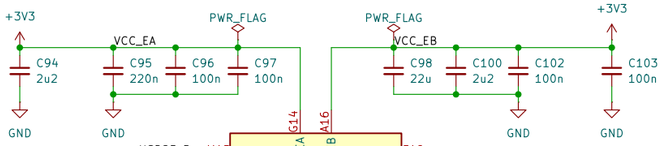
Intergalaktik d.o.o.

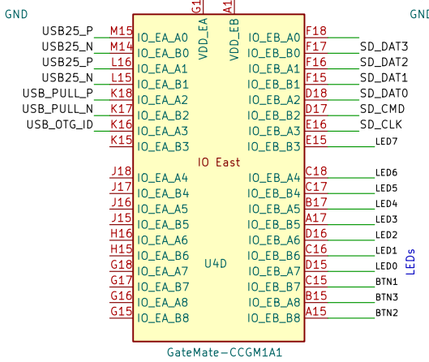
Sheet: /PCIexpress_connector/decap/

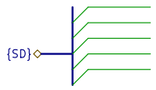
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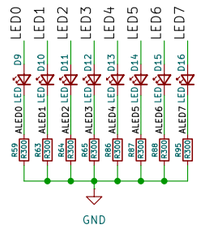
Title: ULX5-PCIe

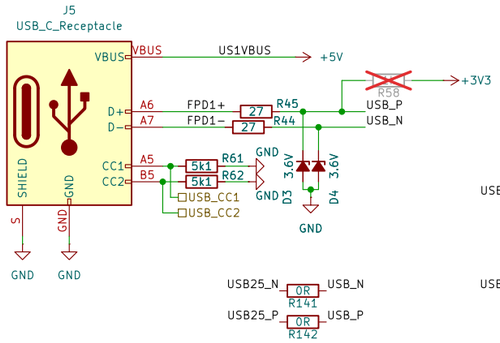
Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 14/11









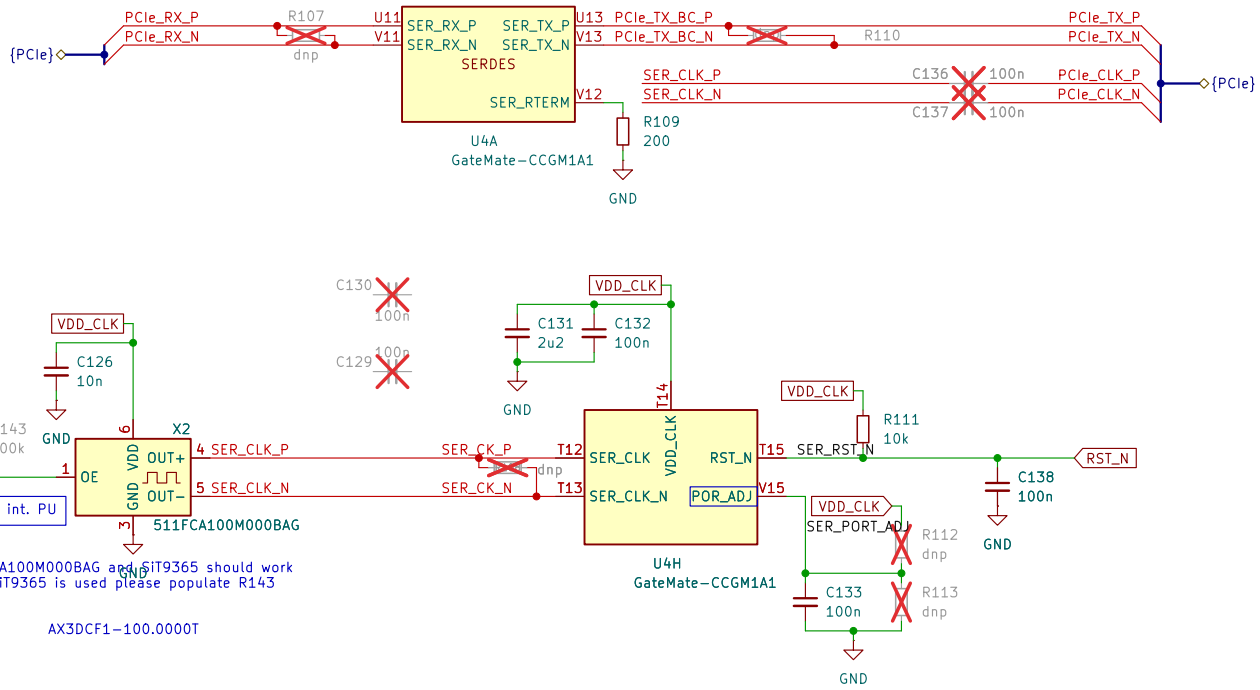




CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /ethernet/
File: ethernet.kicad_sch

Title: ULX5-PCIe

Size: A3	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 9/11



CERN-OHL-S v2
Intergalaktik d.o.o.

Sheet: /serdes/
File: serdes.kicad_sch

Title: ULX5-PCIe

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 10/11